

Wrapup

UCSD ECE 260C Spring 2026

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Agenda

1. “Food for Thought” (re: Seeing Ahead)
2. This Course: ASIC Implementation with Open-Source EDA
3. Context: Age of (System) Optimization
4. Agentic AI and Machine Learning: How Will Semiconductors Ride This Wave?
5. “Scaling of Value” Means Optimization (example: Full-Stack, Low Power)
6. Looking Ahead

“Food for Thought” (re: Seeing Ahead)

Digression on “Big Picture” ... (adapted from CSE 101 lecture)

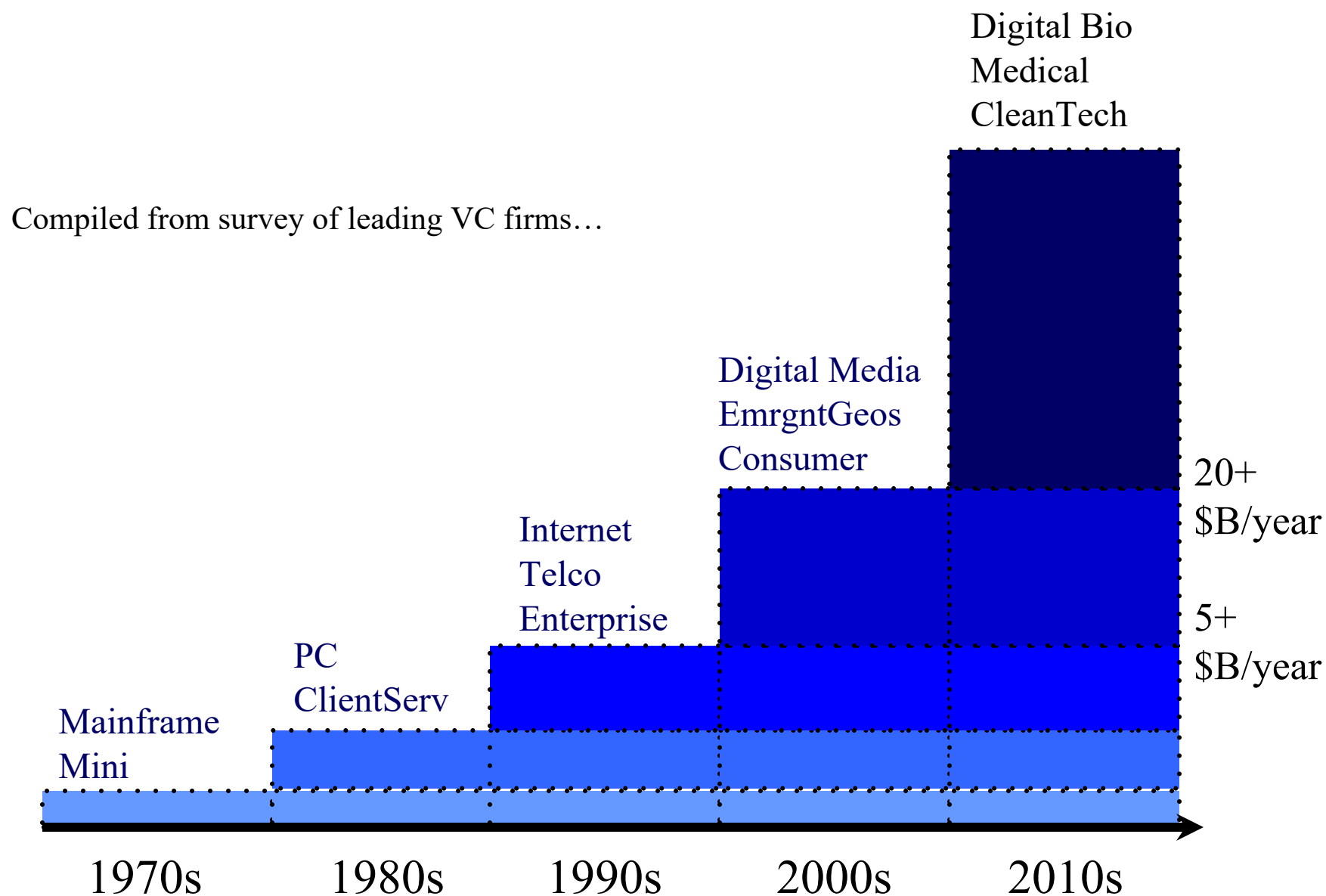
Starting in 2005, I served on Samsung Electronics Corp.’s “Technology Advisory Council” (~8 professors from around the world). The “TAC Forum” each year included a panel discussion to address various questions of current interest to the company. The following “context” slides are semiconductor-centric, but convey the point of “looking at the big picture”.

2006 TAC



Technology Waves and VC Investment

2006 TAC



Source: ITRS Design ITWG, April 2006

1. What Applications Will Lead ?

- Communication and signal-processing applications continue to displace computing applications in defining “paradigmatic ICs”
- Bioelectronics becomes a key application area
 - I.e., all types of applications of integrated electronics to the biomedical area
 - Implantable diagnostic devices
 - Integrated bio-sensors for DNA/protein discovery
 - Cheap medical test instruments
- Leisure
 - Games, entertainment, multimedia, ...
- Automotive, industrial electronics, military, security

2. What Technologies Will Be Required?

- Greater levels of integration
 - Packaging, random-logic and memory, digital and analog, passive and active, electrical and mechanical (including microfluidics), etc. in 3D
- Bio-electronics applications
 - Micro-fluidics
 - Discovering chemical / biological tests that permit detection of reactions by means of optical emission, or thermal or electrical changes in the reagents, in ways that are implementable on chip
 - Issues of cost, integration, and reliability must be understood
- General electronics
 - Require success of very high density self-aligning technologies (nanotubes, molecular transistors)
 - → 3 orders of magnitude density increase, less switching energy compared with end-of-scaling CMOS
 - Must comprehend optimal architectures and even algorithms that exploit very high density of emerging technologies

3. Advice For Students and Engineers?

2007 TAC

- **Students: De-specialize !**

- Be creative and aware
- Learn physics, chemistry or biology
- Learn how to program parallel and heterogeneous systems
- Understand the context for what you do, and why you do it

- **Engineers: Engage !**

- Attend conferences, read, publish
- Formulate what you want researchers to deliver
- Engage with university community

- **VPs and CTOs: Differentiate !**

- People, tools, processes, cultures, IP
- Don't be shy with external research \$
 - Finger on the pulse of innovation and IP
 - Cheap lock-outs / exclusivity of access

2021:
Algorithms
ML
Optimization
Statistics
Parallel/Dist
Python
Systems Eng
“full-stack”
...

		Differentiating Value Added	
		Low	High
Permanence	Easily Lost	Processes	People
	Easily Kept	X	Tools, IP Cultures

- **Continued Moore's Law scaling of integration, value**
 - Three fundamentals: memory, interconnect, 3D
 - Scaling to $< 20\text{nm}$ based on photolithography is **unclear**
 - Equivalent scaling from software, stacking, architecture, ...
- **Trajectories**
 - *Applications* still driven by mobility, bio, clean tech, leisure
 - Portable multimedia, graphics resolution, core network bandwidth
 - **Photovoltaics !!!**
 - *Massive storage* driven by Flash (e.g., TB in cell phone)
 - *Human-machine interaction*: Intelligent, cognitive, natural
 - Augment reality with a virtual world
 - *Power and energy*: harvesting, management, portability
 - *Closing the committed vs. programmable ("FPGA-ASIC") gap* via architectural organization of processing, memory, communication blocks
 - *Circuits* (subthreshold; multi-radio smarts and interfaces) *and systems* (spatially-adaptive, ultra low duty-cycled); *Processing* (driven by parallelizable applications)

- **Bifurcation between developed world and emerging countries**
 - Deepening bifurcation between high-end and low-end semiconductor industries
 - Emerging regions will be increasingly important
 - New global players continue to come from emerging countries → acquire developed-country players
- **More Than Moore**
 - Growing part of product value comes from highly complex software and non-digital/CMOS (analog, MEMS)
 - Must be a major focus for innovation
- **Structural changes due to PV, memory**
 - PV (CZ/FZ/polycrystalline) impact on Si supply chain
 - Imaging / image sensing, mass storage, NV affect memory
- **Emergence of an applications long tail?**
 - Heterogeneous multi-core with programmable fabrics → opens door to the long tail of embedded applications

- **Must be able to deliver SW/HW solutions**
 - Limiting case in 2020: HW is free, consumers buy services
 - Design technology is critical
 - Who will be first to enable parallel SW development?
 - Who will leverage DFM to regain cost trajectory of manufacturing?
- **Global collaboration is key**
 - Sharing of IP, NRE, fab cost, royalties; JV participation
 - Don't be isolated
- **Must have big footprint in emerging economies**
 - Aggressive collaboration and partnership policy, strategy
 - Acquire aggressively to spur growth and innovation that cannot happen enough internally
- **Shift the cost crossover point for low-volume product**
 - Else, IC sockets go unused or filled by expensive silicon
- **Keep an eye on:**
 - Obvious: IP, new memory technologies, ...
 - Energy application space
 - Massive energy capture or dissipation through semiconductors (solar capture, wind capture, thin layers that isolate layers of clothes)
 - Online energy regulation, downstream effects of PV, ...

Takeaways Aligned With This Course (YMMV!)

- **Think Big Picture**
- **Richard Hamming, “You and Your Research”**
 - Each week from Friday noon onward, only think about “big problems”
 - <http://vlisicad.ucsd.edu/Research/Advice/hamming.html>
 - <http://vlisicad.ucsd.edu/Research/Advice/index.html> → [check this out](#) (Pausch on time management, Raibert on good writing, ...)
- **Steve Teig**
 - In 2002, broke Edison’s record for number of patents filed in a single year by an individual
 - 1997 conversation about “things to work on”: [turn cellulose into gasoline, recover gold from seawater, ...](#)
- **Build transferable skills**
- **You will still be working 50 years from now. What skills should you learn today?**
- **Toolkits:** global optimization; machine learning; concurrency; systems design; modeling and simulation; ...
- **Skills for the engineering life:** how to study for a lifetime; how to be flexible; how to stay curious, inventive and motivated; ...

This Course: ASIC Implementation with Open-Source EDA

- Hardware design is inseparable from design tools and optimization mindset
- Knowing what tools think internally, and what they think about – and that you can change the tools (!) – has value
- Agentic AI / Big Data is coming for every vertical, including semiconductors
- Full-stack optimization has biggest levers at the top
 - “Last levers”: quality, schedule, cost
 - At the same time: hardware design is “different and difficult”

Basics: What, Why, How, Where, What If ...

- Where do I learn about the state of technology?
- **What are the major challenges to chip implementation?**
 - How much does chip implementation matter?
 - Which challenges are getting worse? better?
- **Explain how electrical parameters of devices and wires are scaling**
 - Gates
 - Wires
 - Dynamic power, leakage power
 - Speed – across temperature, voltage
- **What gets worse, and why?**
 - Variability, Parasitics, Leakage, Reliability, Power Density, Thermal Gradients, Stress, Coupling/Noise, ...
 - What kinds of variability?
 - What kinds of reliability
- **Explain major sources of variability**
 - Systematic, random
 - Polygon-level, die-level, beyond-die (wafer, lot, ...)
 - Litho, implant (FEOL)? CMP, litho (BEOL)?
 - Are these getting worse or better in future technology nodes? Why?
 - How do we model, manage, mitigate today?

Basics: What, Why, How, Where, What If ...

- **What are tools thinking when they run?**
 - Floorplanning
 - Global placement
 - Detailed placement
 - Clock tree synthesis
 - Global routing
 - Detailed routing
 - (Physical verification, DFT, ...)
- **What are the key algorithmic frameworks?**
 - Placement: Annealing, force-directed, recursive bipartitioning, and “hybrid” methods
- Objective functions?
- Underlying models of the design/netlist?

Basics: What, Why, How, Where, What If ...

- **What converges throughout the chip implementation flow?**
 - Parasitics, spatial embedding, timing and power
- **Why do we perform floorplanning / powerplanning, placement CTS, global route, detailed route, ... in this order?**
 - (Converging of spatial embedding and parasitics)
 - Why is a “one-shot, convergent” flow important? (productivity)
 - Why is pessimism important? (avoid loops)
 - What are three ways of “predicting” downstream flow stages?
 - (1) Constructive prediction; (2) Models and statistics (e.g., wireload models; table models); (3) Assume and enforce; ...
- **Where are there loops in the implementation flow?**
 - Placement and clock tree synthesis
 - Netlist and physical design
 - ECO P&R and sizing
 - Physical design and timing design
 - How do I break or manage these loops?
- **Where are there loops in the analysis flow?**
 - What is the loop involving (routing), crosstalk and timing?
 - What is the loop involving power, temperature, timing?
 - What is the loop involving aging, voltage scaling, power, temperature?
 - What is the loop involving leakage, temperature?
 - What are loops involving reliability? (EM, NBTI, TDDB, HCI, ...)
 - How do I break or manage these loops?

Ideas: Models and Objectives

- Wiring models
 - Before design: Rent's rule: Rent parameter p implies a wirelength distribution
 - Before design: Previous chips
 - Before synthesis or floorplan: generic, block-specific wireload model
 - During global placement: HPWL, Steiner tree with layer assignments, ...
 - Spanning tree, single-trunk Steiner tree, ... (why?) (what pros and cons?)
 - ...
- Parasitic models
 - Used where? R , C , L , ... coupling, grounded, with vias / without vias, above the cell, including the cells, ...
 - To capture what?
- Delay models
 - NLDM / Liberty (per arc, per transition table-lookup)
 - C_{total}
 - $C_{effective}$
 - Elmore delay
 - MCF for C_c
 - Why is pessimism important? (What are the costs of pessimism – guardband reduction example)

Optimizations

- Which optimizations are getting relatively easier vs. harder?
- Which optimizations are “new” or “important in the future”? Which knobs are important?
 - Floorplan, power plan (voltage ...), clock design, ...
- Dynamic power reduction
 - Clock gating
 - Voltage scaling
- Leakage power reduction
 - MTCMOS
 - Lgate biasing
 - Power gating

Context: Age of (System) Optimization

SCAS-2022 keynote video: <https://www.youtube.com/watch?v=5k-Hf0pUjU>

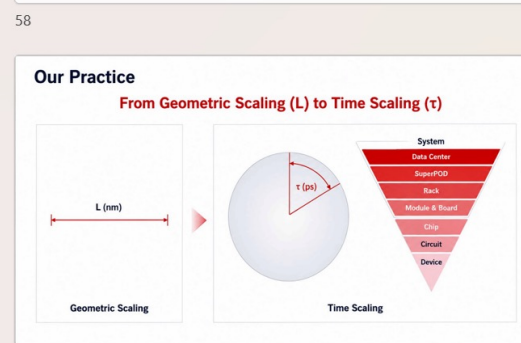
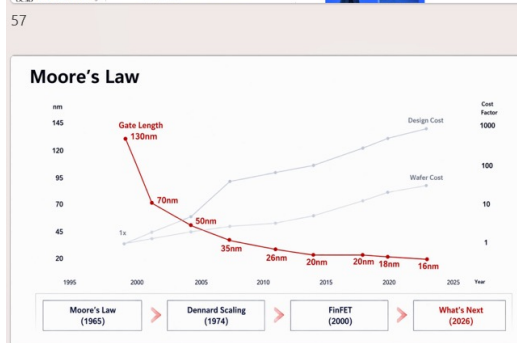
4. Huawei and the “Tau Scaling Law”

New Semiconductor Path in Practice

τ Scaling Law for the Post Moore Era

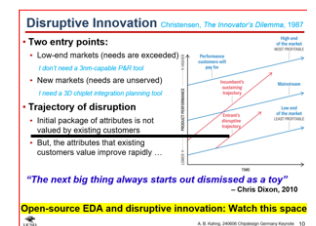
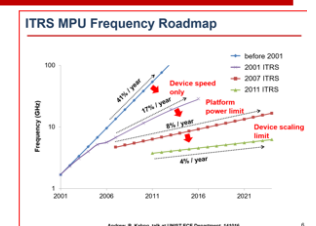
Tingbo He
President of Huawei Semiconductor
Chair of Huawei Scientist Committee

May 2026



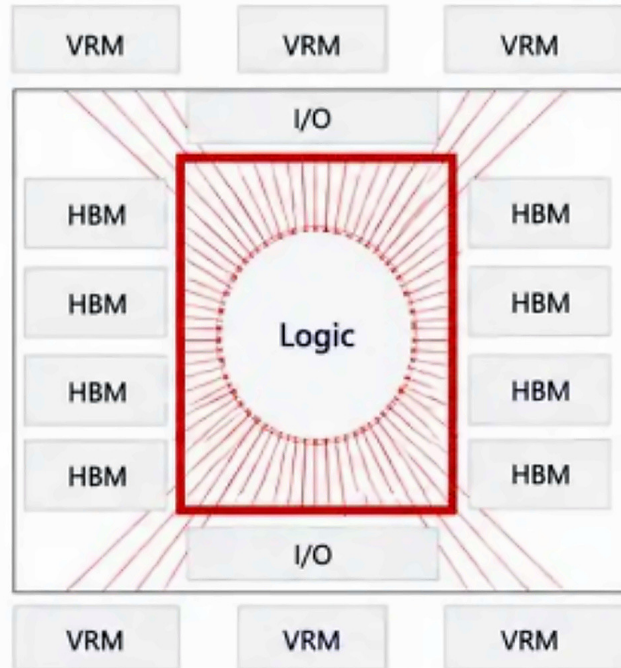
Think for Yourself *(if only because others are thinking hard ...!)*

- The Innovator's Dilemma
- Necessity is the mother of invention
- Five questions
 - How should a general business audience understand Huawei's "Tau Scaling Law"? If chip progress is no longer driven only by shrinking transistor dimensions, what exactly is being optimized or "scaled"?
 - From an EDA and physical-design perspective, how important are shorter signal paths, better layout, interconnect optimization, and design-technology co-optimization for continuing chip performance improvements in the post-Moore era?
 - As traditional lithography scaling becomes more difficult and costly, to what extent can system-level design, advanced packaging, 3D integration, and hardware-software co-optimization help extend the performance and energy-efficiency gains of semiconductors?
 - Huawei has referred to a future capability "equivalent to 1.4nm." From a design and implementation perspective, how should such an "equivalent" node be understood, and what kinds of measurable results would be needed to support that claim?
 - If Tau Scaling or similar approaches prove successful, what positive impact could they have on AI chips, data-center computing, chip design automation, and the broader transition beyond Moore's Law?



System Folding to Overcome 2.5D Scaling Dilemma

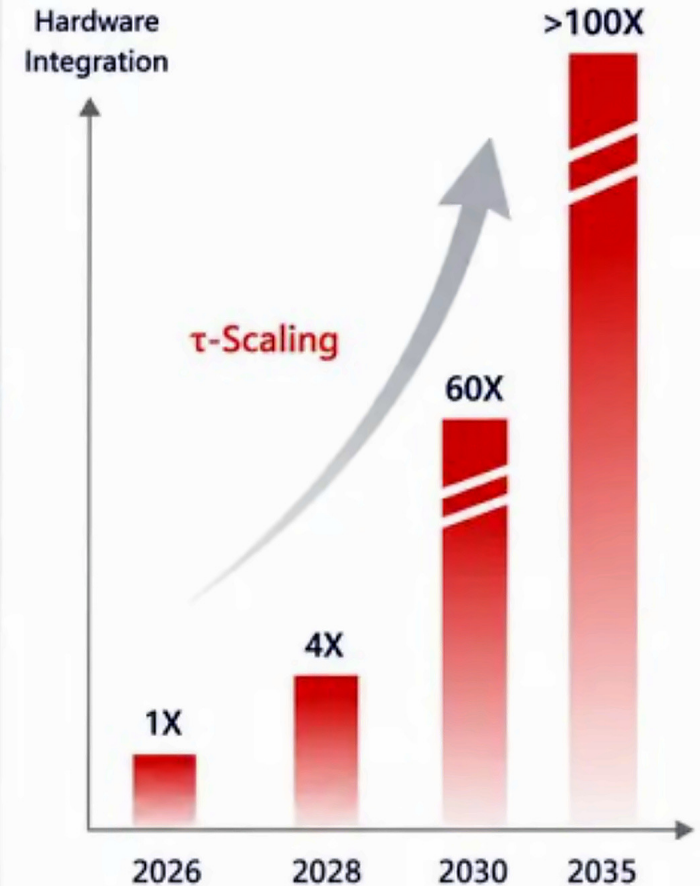
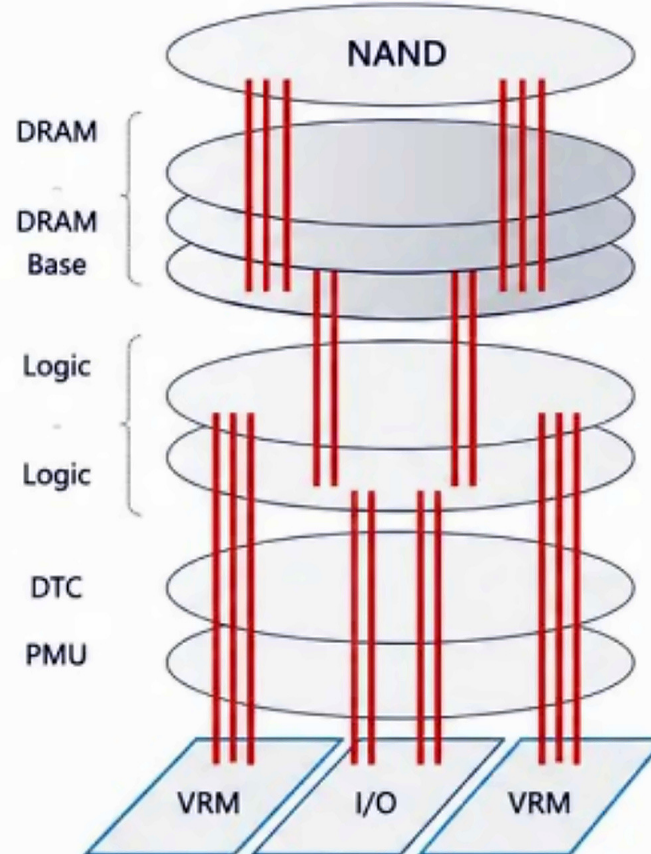
2.5D Dilemma



N^2
Die Size
Performance
Power

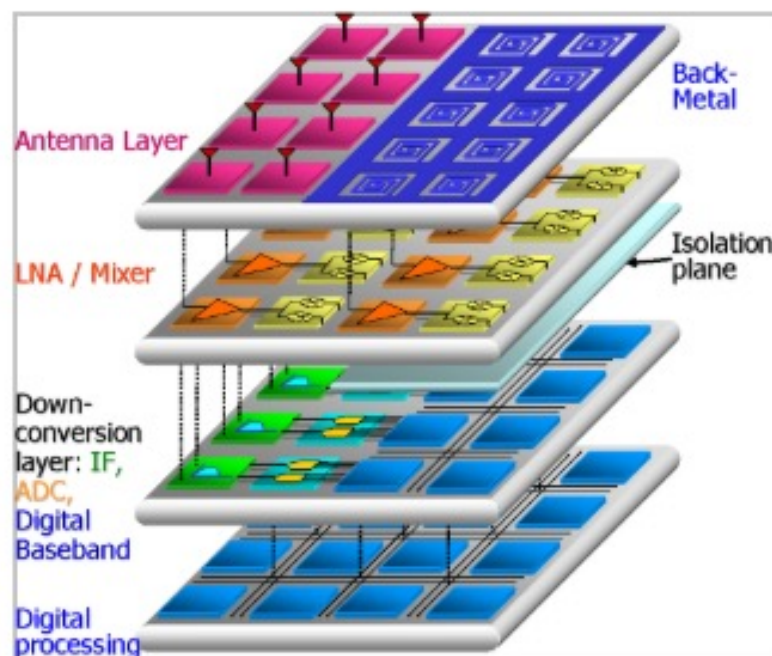
N
Memory BW
Interconnect BW
Power Delivery

System Folding

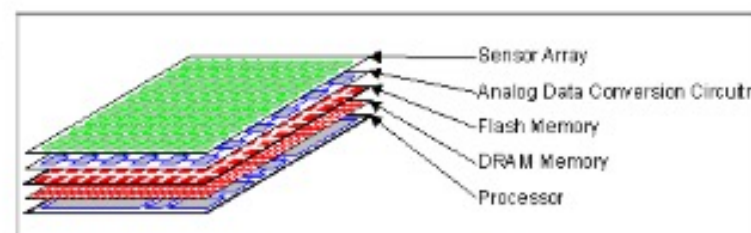


ECE 260B / CSE 241A, 15 Years Ago

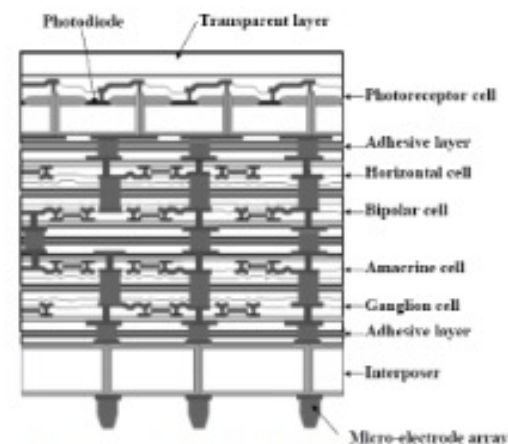
Example 3D Systems



[K. Bazargan, R. Harjani
University of Minnesota]



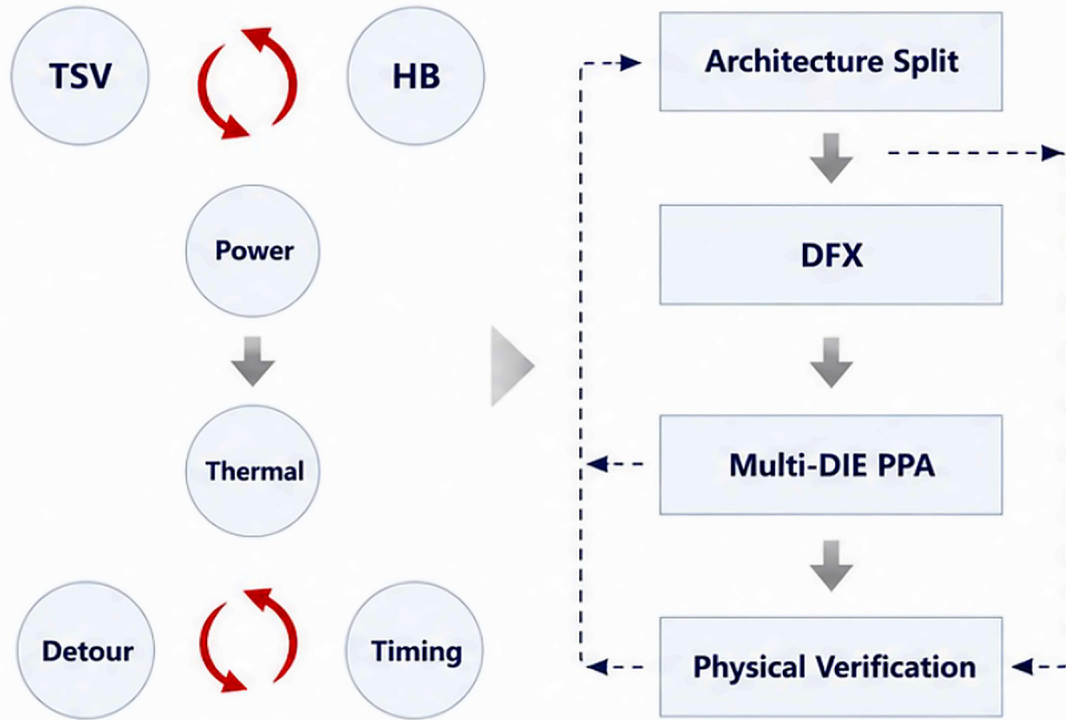
[www.tezzaron.com]



[Koyanagi, Tohoku U./Zycube]

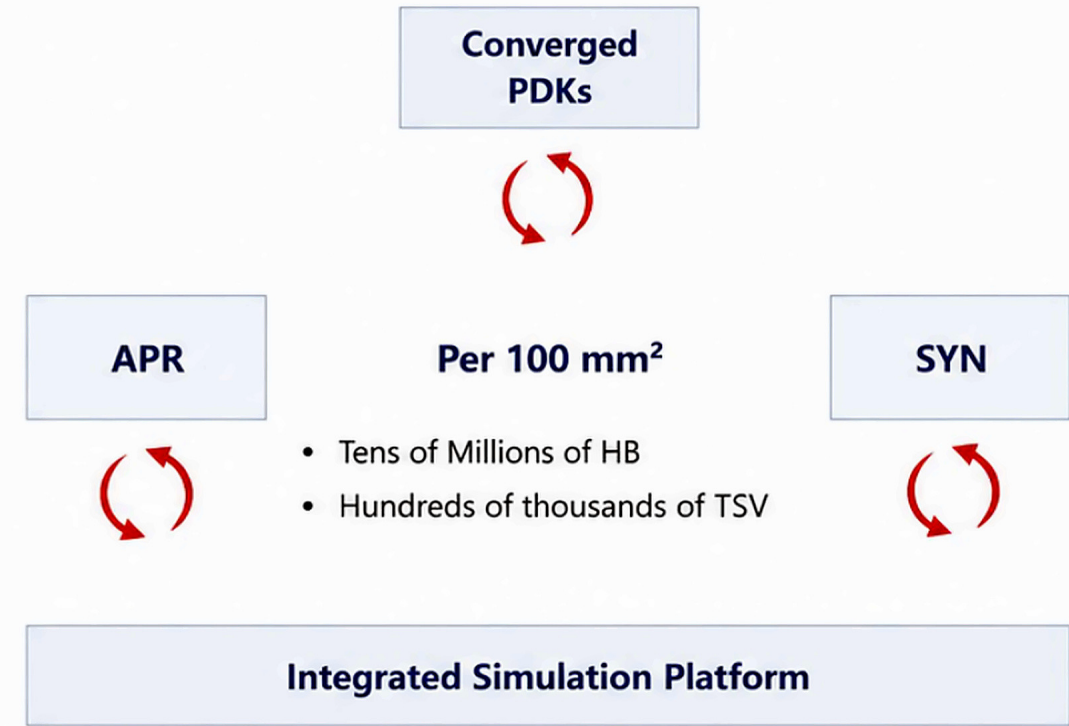
Folding-Oriented Methodology and Tool

Methodology



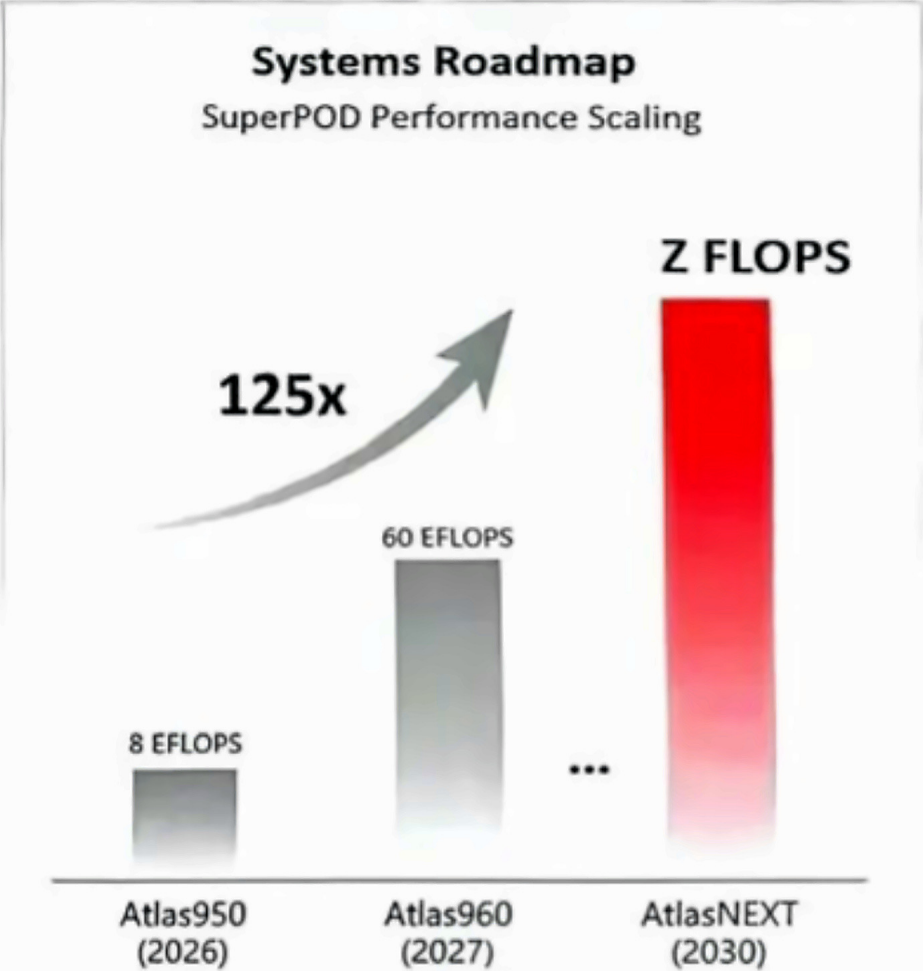
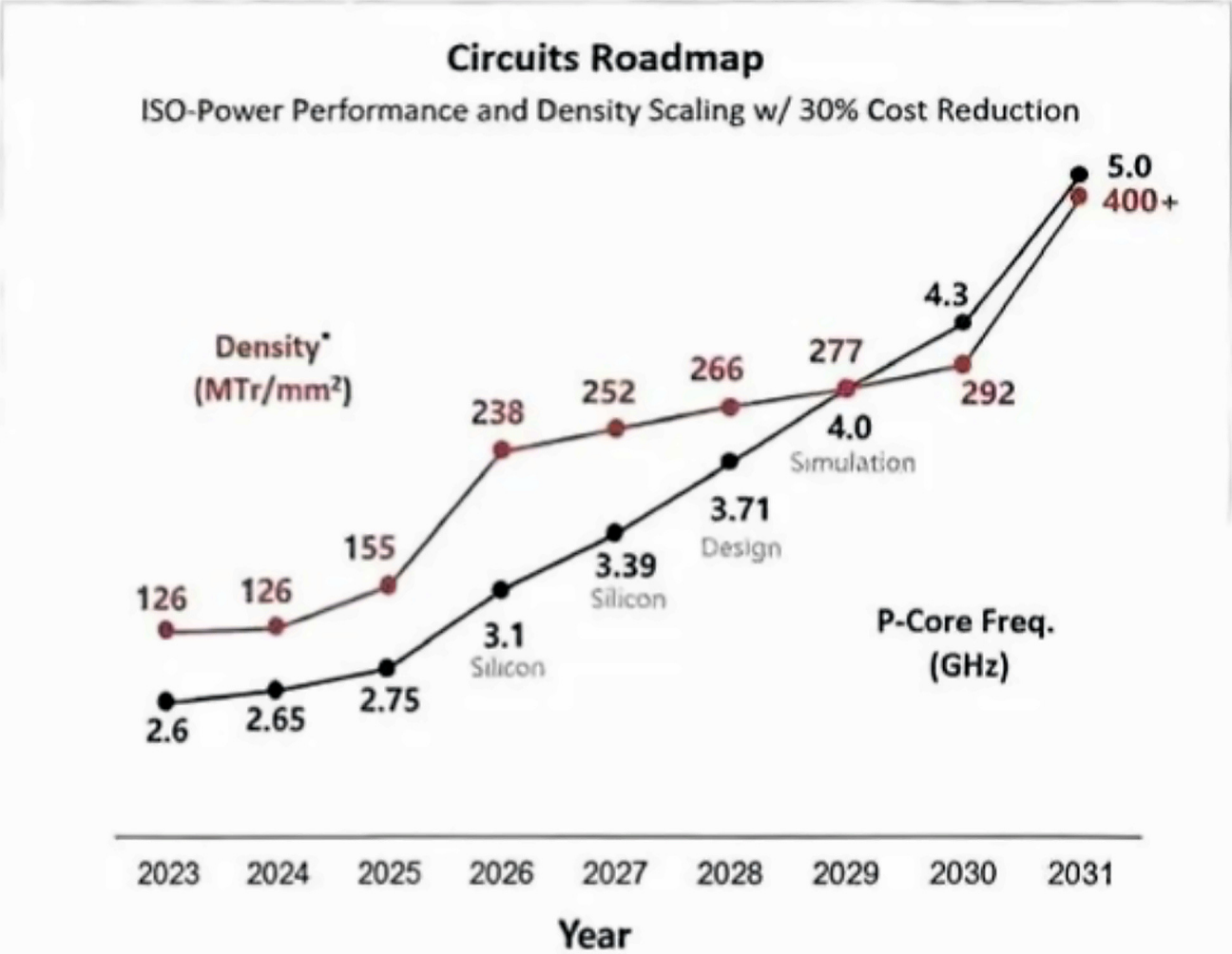
Concurrent Optimization

Tool



Intelligent and Efficient Platforms

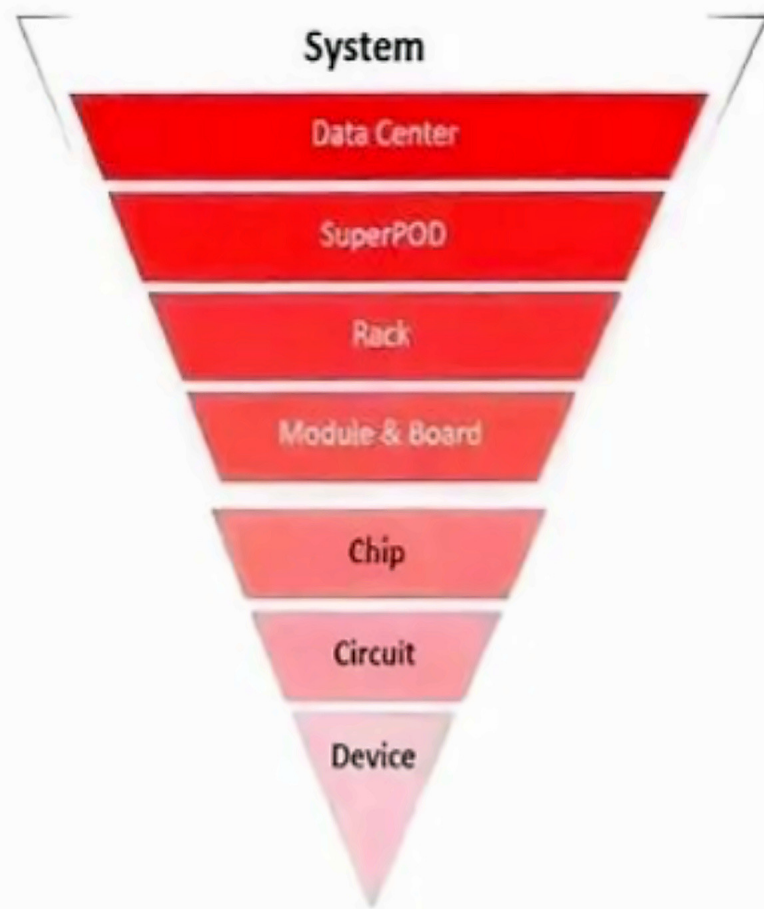
τ-Scaling Roadmap: Sustainable PPDC Evolution



Note*
 $Density = \frac{2}{CPP \cdot cell\ height}$, $D_{Design} = Density \cdot Design\ Utilization(60\sim75\%)$

τ -Scaling: Universal and Sustainable Law

$$\tau = f(\tau_{tr}, \tau_{circuit}, \tau_{chip}, \tau_{sys})$$



Level	Objective Function	Temporal Scale
Device	$\tau_{device} = (R_{gate} + R_{net}) * (C_{gate} + C_{load})$	ps
Circuit	$\tau_{circuit} = f\{\sum(\tau_{transistor} \& \tau_{interconnect}), DTCO\}$	ns
Chip	$\tau_{chip} = f\{\sum\tau_{cir_{XPU}}, \sum\tau_{cir_{MEM}}, \sum\tau_{cir_{IO}}, N_{parall}, \eta_{eff}, M_P\}$	μs
System	$\tau_{sys} = f\{\sum\tau_{chip}, \sum\tau_{comm}, N_{parall}, \eta_{eff}, M_P, RAS, STCO\}$	ms ~ s

Agentic AI and Machine Learning: How Will Semiconductors Ride This Wave?

Toward an open vertical environment for EDA agents

Environment scaffolding (The substrate agents act inside)	Agent capabilities (What we want agents to do)	Ecosystem and governance (What OpenROAD can achieve)
Episode schema for EDA (workspace, actions, constraints, QoR)	Lexicographic, constraint-aware rewards (hard gates → QoR frontier → cost)	Open evolving agent benchmark suite
Cross-stage state & observability (RTL ↔ verif ↔ synth ↔ PD ↔ signoff causality)	Cross-tier orchestration (flow-level + code-level in one loop)	Reproducibility manifests (pinned commits, PDKs, agent versions, prompts)
Multi-fidelity ladder (lint → sim → synth → PD → signoff audit)	Trajectory-grounded training + research-grade memory	CI-enforced QoR gates + machine-checkable invariants
Fast learned surrogates as first-class env citizens	Code-level R&D	Industrial calibration loop + compute equity

What four systems (**HELIX, (GR + ABC)-Evolve, MAGE**) agree on

Grounding + physical feedback

- Domain grounding and physical feedback are complementary; not substitutable
- The agent must know the domain and see the physical consequence of its edits

Correctness guards

- Correctness guards are what make autonomous evolution economically viable
- Without automatic gates, you waste resources on invalid states and false QoR

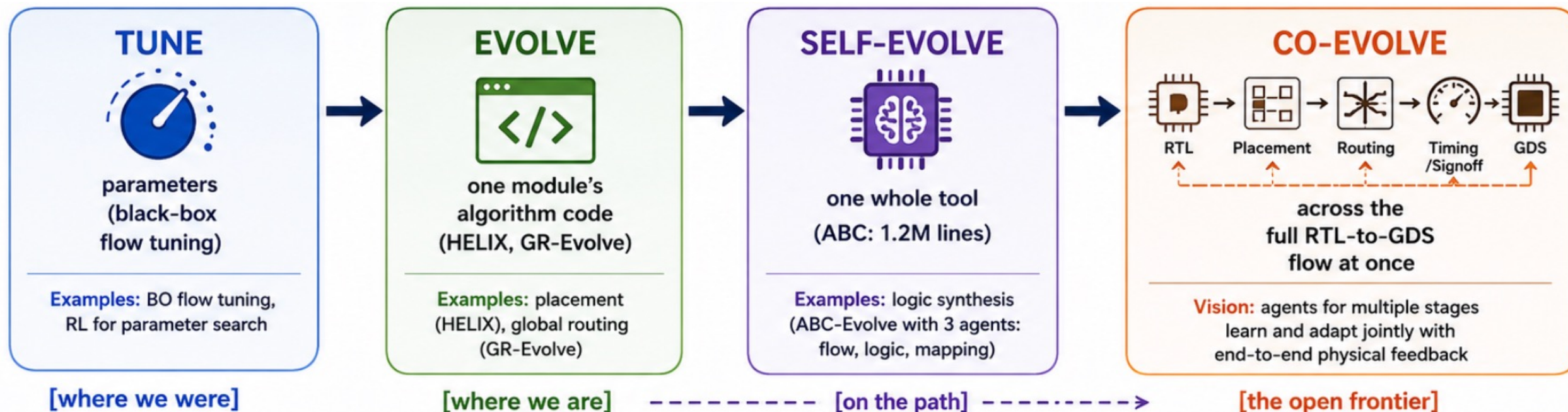
The loop matters

- The evolutionary loop, not raw model capability, drives the result
- Better models help, but the scaffolding around it does the heavy lifting

Agents amplify priors

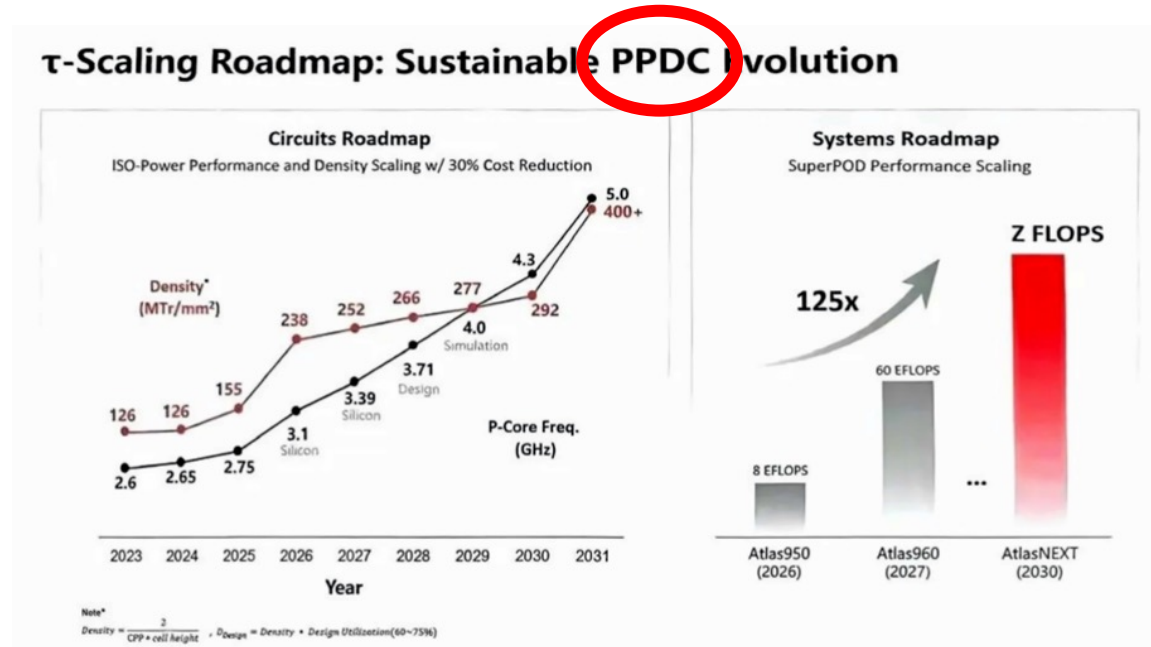
- Today's agents amplify human priors – they do not yet invent paradigms
- Win is a force-multiplier on human-curated knowledge, not autonomous invention

The spectrum: where we are heading



The big picture: *Agentic PD systems are scaling in scope. The frontier is full-flow, multi-objective, and jointly optimized. We're not there yet – but the path forward is clear.*

“Scaling of Value” Means Optimization (example: Full-Stack, Low Power)



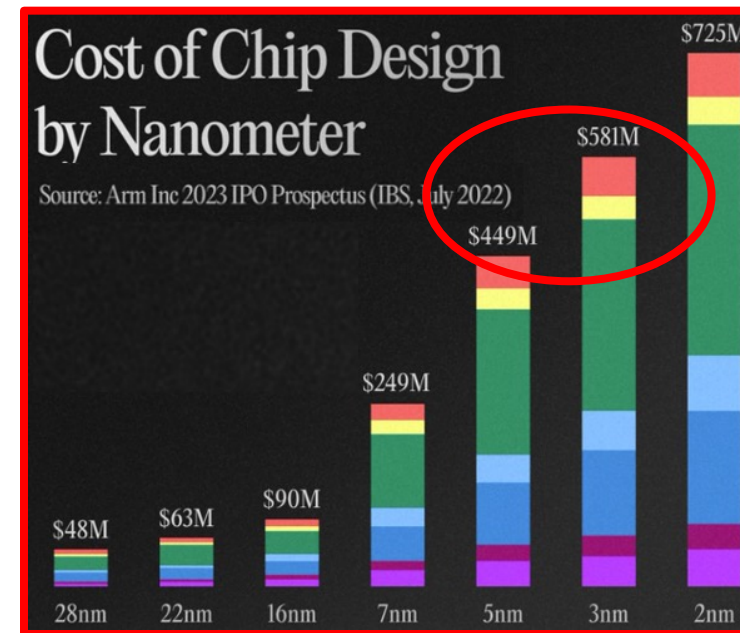
How Much Does it Cost to Design a Chip Today?

- A. \$10M
- B. \$30M
- C. \$100M
- D. **\$500M**

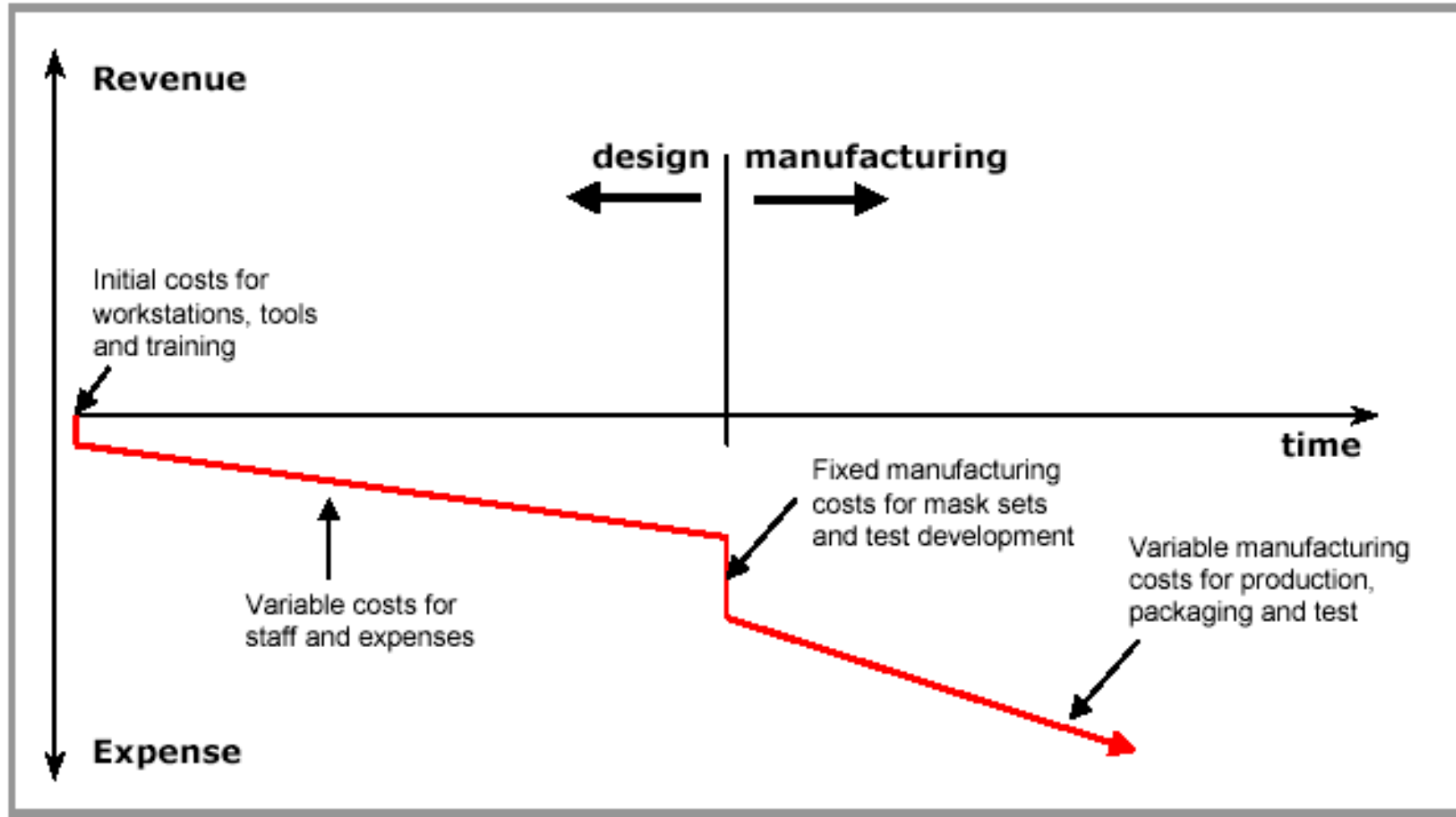
“A 65nm chip in 2006 cost about \$30 million to design, while **a 5nm chip in 2020 cost over \$540 million to design.**”

-- Semiconductor Industry Association

The Problem: Cost of design blocks hardware innovation !



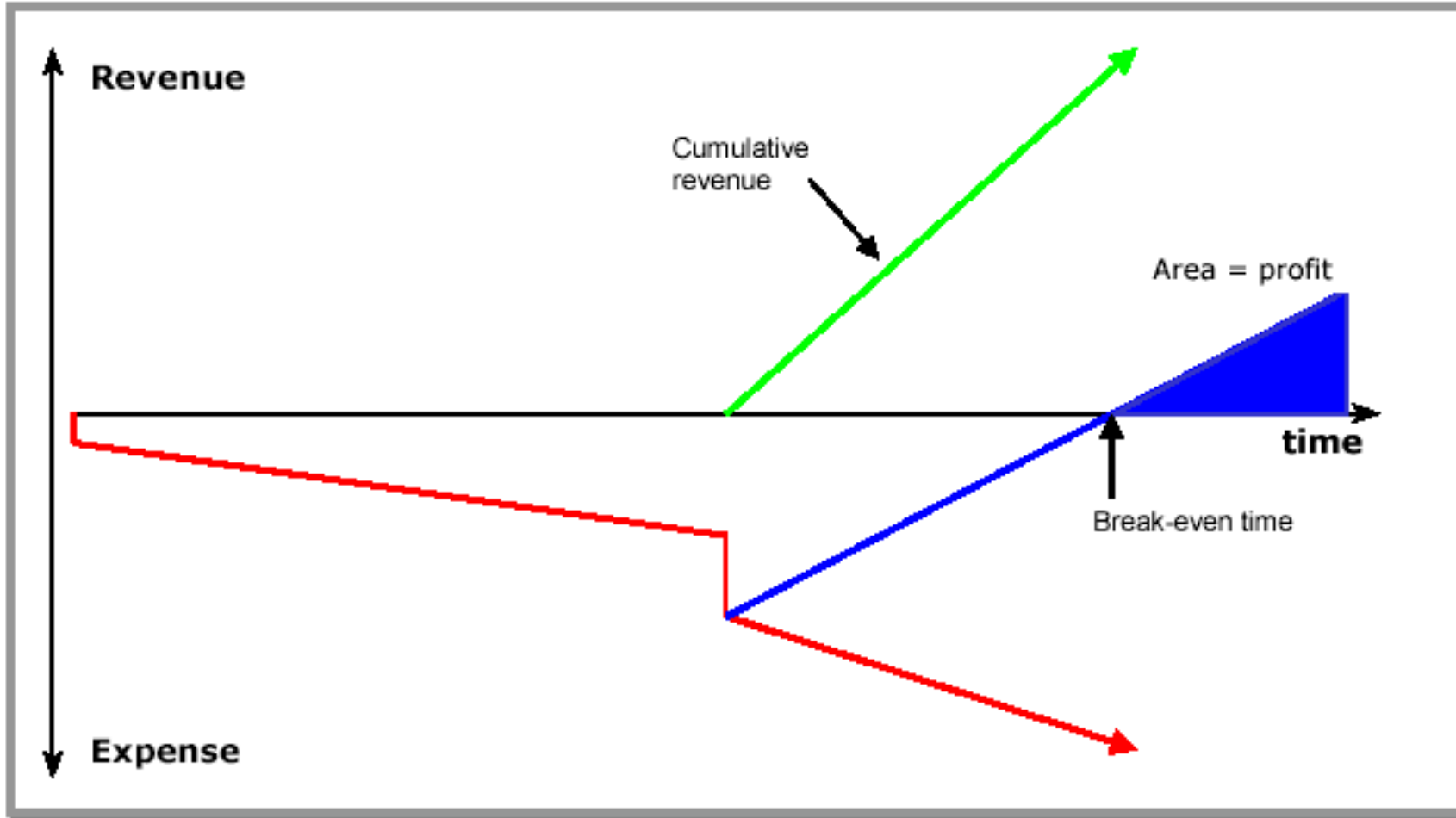
Idealized Cost Curve



Source: Nassda Corporation

Revenue and Profits

- Cumulative profit = cumulative revenue – cumulative cost



Source: Nassda Corporation

Example: Low power implementation: Why Power Matters ?

- Power Consumption is key differentiator factor for :
 - High performance computing (HPC)
 - ML and AI devices operating in the cloud or at the edge
 - XR, VR devices (Smart glasses)
 - Automotive devices (SOC's used for self driving cars, Infotainment, Communication)
 - Mobile devices: Cell phones, Wireless gaming consoles
 - IoT: Sensors, Wireless Headphones
 - Consumer electronic compute devices: Laptops, tablets
 - Medical devices: Implantable medical devices (pacemakers, defibrillators, glucose monitor devices)
 - Drones
 - Toys

Low power implementation : Metrics

- User experience perspective (marketing specification):
 - For mobile devices:
 - Active time of the device: Time interval of performing a well-defined set of tasks (defined use mode: audio play , voice call, web browsing, video playback, etc) between two battery charges
 - Standby time of the device: Time interval between two battery charges when the device is fully functional ready to be activated but does not perform any user driven tasks.
 - Current at the battery during the execution of a benchmark test
 - DOU (days of use) – key metric (KPI) for comparative analysis
 - RBSC (rock bottom standby current) per power domain
 - For electrically powered devices:
 - Efficiency: Power consumption for performing a defined set of tasks relative to performance metrics: mW/Mhz for processors and DSPs (or custom defined figure of merit, TFLOPS/mW)
 - Average and Peak power consumption
 - System power functions:
 - $P(f, \text{bandwidth, MIPS (million of instructions per second), } n:\text{system parameters, } t)$
 - Adaptability: Capability of the system to self-adjust its workload and power consumption relative to peak currents, process, temperature and voltage variation

Low power implementation : Metrics

- Power, performance and area metrics (PPAC) are driving the design requirements for low power implementation:
 - System level
 - Architectural level
 - Software implementation
 - Hardware implementation (IC design)
 - IP design (Standard cells library, Memories, DDR, PLL's, USB3.0 etc)
 - Manufacturing (process technology)
- Power consumption in digital systems:
 - $P_{\text{total}} = P_{\text{active}} + P_{\text{leakage}}$
 - $P_{\text{active}} = P_{\text{internal}} + P_{\text{switching}} = P_{\text{internal}} + \alpha C V^2 f$

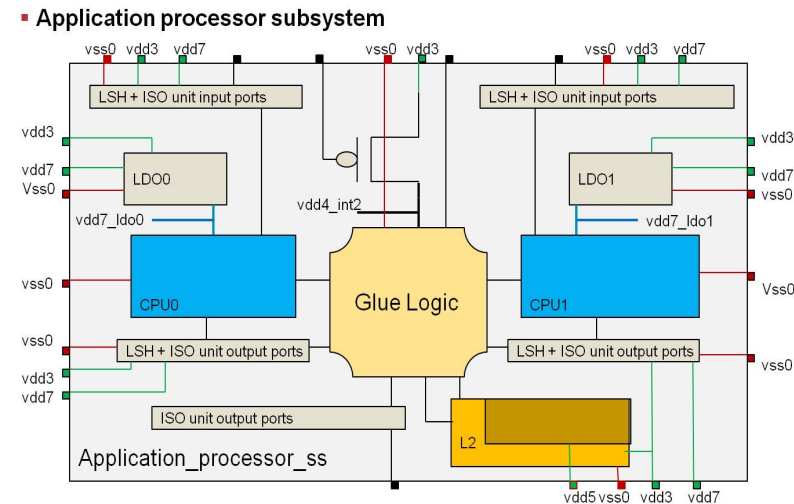
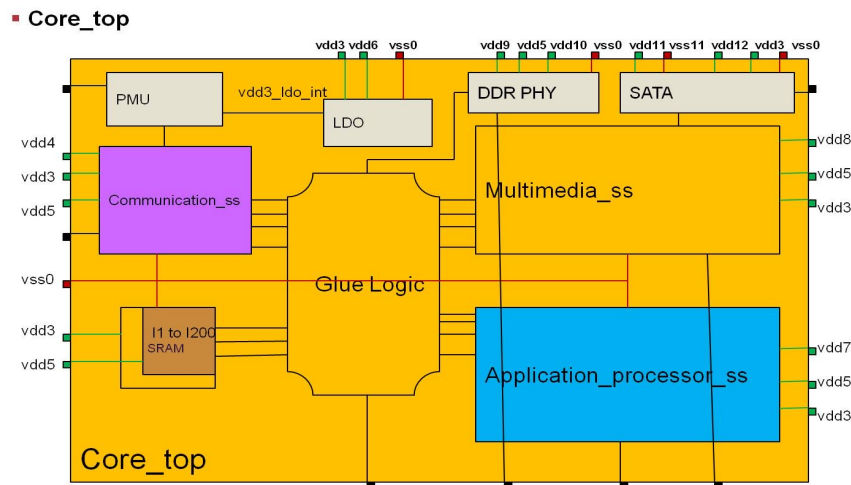
V – voltage
f – frequency
C – capacitive load
 α – activity factor

Low power implementation: Design synergy

- Low power implementation for the modern system on chip (SOC) requires a holistic and concurrent approach which includes collaboration between:
 - System level design (Subsystems definition , PKG, PCB, System PG/PDN)
 - Architectural design.
 - Software optimization and SW-HW co-design
 - Power aware RTL imp./synthesis/optimization (front end design)
 - Physical design (chip/subsystem/block level) (back-end design)
 - IP design:
 - Circuit design, Physical implementation of the IP
 - Process selection (Bulk CMOS, SOI, FinFet, GAA) and device definition (nmos, pmos). Design Technology Co-Optimization.
 - Adaptive design (on die sensors, process aware voltage scaling, process, power and temperature monitors). Power models (Upf)
 - Power and thermal design ,verification and modeling (DFP)
 - Design for Test (DFT), Manufacturing (DFM) and Reliability (DFR). Silicon char., power models validation, silicon to model correlation,

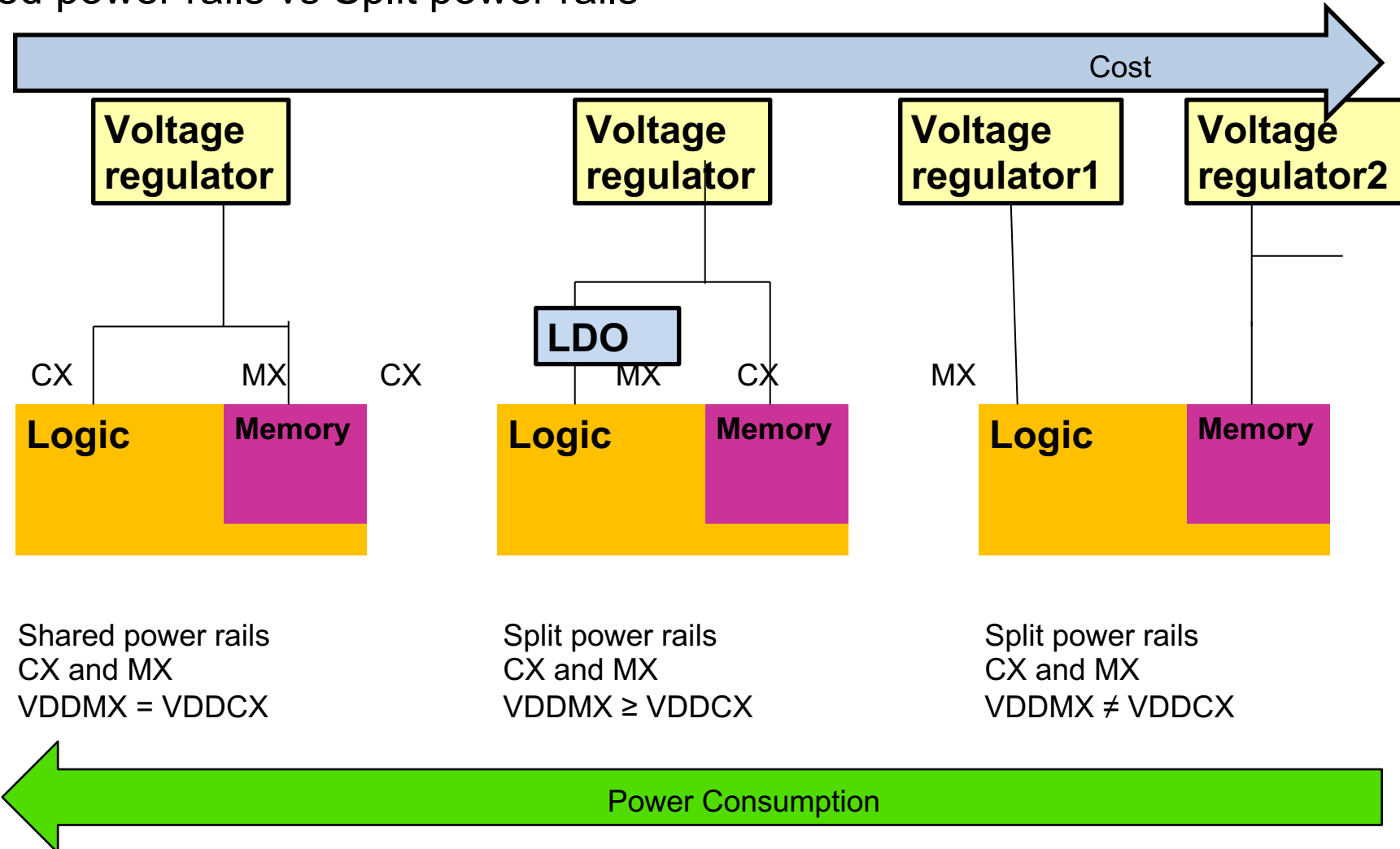
System level optimization

- Power delivery network optimization (PDN):
 - On die vs. on board (PCB) voltage regulators
 - Voltage regulators efficiency
 - Voltage rails definition (always on, collapsible, shared power rails, split rails)
- System level power management techniques:
 - Adaptive voltage scaling (AVS), Adaptive Power MUX (APM), Power gating
 - Dynamic clock frequency and voltage scaling (DCVS)
 - Static voltage scaling (SVS). (Performance mode definition: Nominal, Turbo, LowPower, NTC).
 - Retention modes/states
 - Limits management (peak currents management, thermal management, adaptive clock distribution ACD)
- Power management architectural implementation



System level optimization

- Power delivery network optimization (PDN):
- Example: Shared power rails vs Split power rails

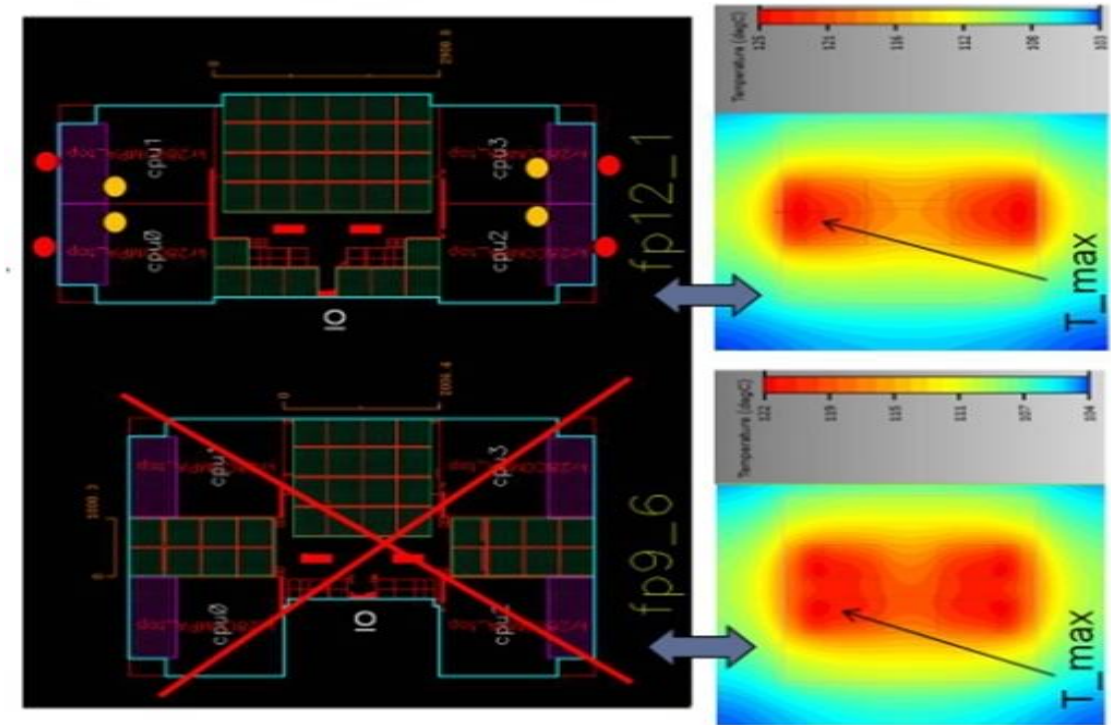


System level optimization

- **Cost versus functionality and user experience optimization for specific target markets:**
 - Process Technology selection:
 - Manufacturing process (process technology: 40nm, 28nm, 20nm, 22nm FDSOI, 16,14,12,11,10, 8,7,5,4,3,2 nm FinFet etc.)
 - Die area and cost (cost of silicon, number of good dies per wafer, wafer cost, mask field utilization MFU)
 - Package selection (size, type, suppliers, number of pkg layers)
 - Printed circuit board design (PCB)
 - Power and performance specification for well defined use cases
 - Reliability specification (consumer market, automotive market)
 - Thermal specification
 - Power targets (TDP spec, DOU spec's)
 - Standards and regulation compliance (automotive, healthcare)
 - Defects per million (DPPM)

System level optimization

- Thermal management
 - Design for thermal:
 - Casing, cooling system, PCB – package - die co-design
 - Software development for thermal and power management



- Analog vs. digital processing system level optimization:
 - Optimization at the system architecture with the goal of moving most of the signal processing (data transformation) in the digital domain. The power consumption in the digital domain is scalable with the process technology scaling and with the system use mode requirements.
 - Digitally assisted analog processing
 - In memory computing (ML)
- 2.5D and 3D integration

Architectural optimization

- System on Chip and Subsystem architecture
- Power management architectural definition (power manager)
- Logic to power partition (power domains definition):
 - Always on domains
 - Collapsible domains (power gating)
 - Voltage scalable domains, performance modes
- Power architecture:
 - Functional units, Power buses, Interfaces, Protocols
- Performance levels definition: Level0, LevelU1-U5, LevelD1-D10
- Design Scalability (System on chip – subsystem “plug and play” integration)
- CPU/DSP implementation vs. Dedicated hardware (hardware accelerators)
- Latency, Bandwidth, Throughput
- Software architecture, SW drivers, Firmware implementation
- Software reusability

Architectural optimization

- Memory hierarchy
 - On die vs. off die memory
 - Cache size (miss penalty)
 - Cache hierarchy (architecture, L1, L2, L3 shared)
 - Address space definition , ECC memories
- Processor architecture
 - Von Neumann , Harvard
 - CISC, RISC, VLIW (high IPC),
 - 16bit, 32bit, 64bit, 128bit instruction architecture (IA) (Code compression)
 - In order vs. out of order execution
 - Superscalar implementation
 - Multi thread implementation
 - Scalability : Single core vs. Multi core vs Heterogeneous cores, Big-Little architecture, Power and Performance cores
 - Application specific Instruction Set Architecture optimization
 - FFT cores
 - Multipliers, adders ,shifters

Architectural optimization

- Hardware accelerators:
 - Graphic 2D, 3D
 - Video encoder/decoder (720p, 1080p)
 - Audio + DSP (digital signal processing unit)
 - Modem baseband, Neuronal System Processing (NSP)
- Bus architecture
 - AHB implementation (Advanced high-performance bus)
 - AXI
 - Fabric (high speed, high bandwidth interconnect):
 - Bandwidth
 - Latency
 - Power
 - Network on Chip (NOC's)
- Clocking architecture:
 - PLL's (type of PLLs, number of PLLs)
 - Frequency planning
 - Clock architecture
 - Synchronous vs. asynchronous interfaces

Architectural optimization

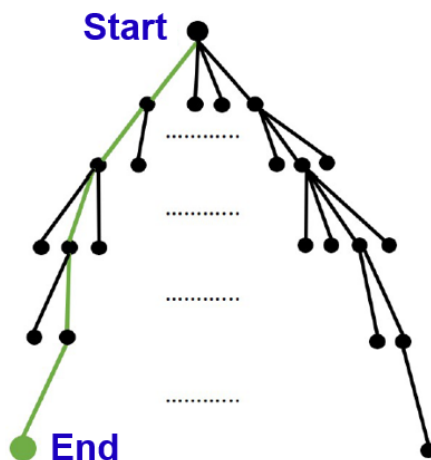
- IO interfaces and peripherals
 - DDR (LPDDR), SDIO, IC2
 - PCIE, USB, MIPI, HDMI, SATA, CLINK (custom link interface)
 - Number of IO pins
 - Throughput and power
- Engineering system level design and optimization (ESL):
 - Algorithmic driven hardware implementation and optimization
 - System level power modeling
- Hardware software co-design and optimization:
 - OS support
 - Drivers development
 - Hardware vs. software control
- Virtual platform:
 - Performance and power estimation
 - Emulation

Looking Ahead

Challenges = Futures = Differentiators

- See farther ahead
- See a bigger picture
 - What are we missing?
 - What are **others** missing?
- PPAC → PPAX (cost, reliability, flexibility, ...)
- “Pathfinding”

Design Optimization Lives in a Box

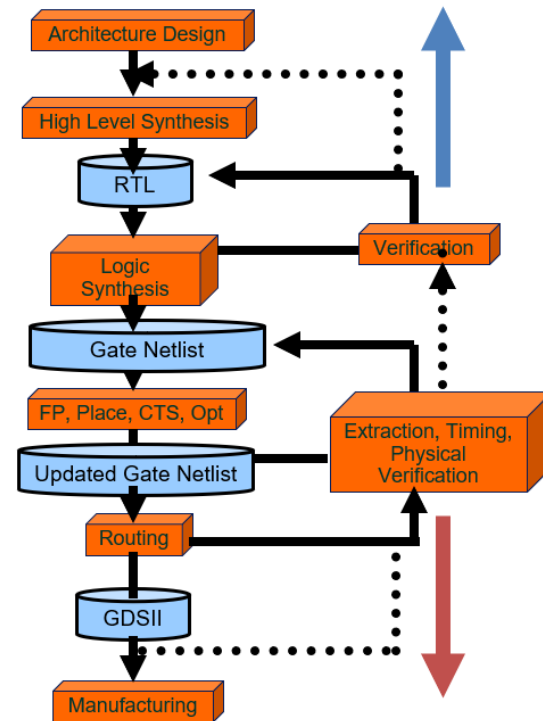


- **Start to End**: expensive!
 - O(year) for product
 - O(weeks) for SP&R and Opt.
- Goal: best possible **End**
- Constraint: stay in “Box”
 - {compute}
 - X {licenses}
 - X {people}
 - X {weeks}

Huge space of trajectories: architecture, enablement, IPs, tools, manual fix, ...

Concept: How the IC Design Flow is Evolving

- Flow expands in two directions
 - System-Level Design
 - Design for Manufacturability (DFM)
- More design care-about
 - Area, Timing, Power, Signal Integrity, Power Integrity, Reliability, Cost
- Key challenges: loops, chicken-egg
 - “Design closure” through tight integrations
 - RTL, GDSII “signoffs” = business structure of semiconductor creation
- “One-pass flow”: required for Productivity, requires Predictability
 - By Guardbands?
 - By “Unifications”?
 - By Statistics?
 - By Methodology to avoid problem?



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Looking Ahead

- **“Where will I use this?”**
- **“What did I learn?”**
- **“How does this apply to my career?”**
- **“What skills do I need to be aware of / what dots must I connect?”**
- **“How can I better understand challenges of getting/staying ahead?”**

- **My hope: that you will avoid “*I wish I had ...*” 15 years from now !**